

Profile

Backend Engineer with 5+ years building **reliable, high-performance services** for critical financial services platforms. Hands-on with **Java, Spring Boot and Spring WebFlux** for reactive, non-blocking microservices, and with **RESTful and event-driven APIs** built on **Kafka and RabbitMQ**. Experienced designing **resilient distributed systems** with **idempotency, eventual consistency and fault tolerance** at the core, integrating banking and mobile money systems that process real-time transactions. Strong advocate for clean code, test-driven development, and automated CI/CD, working closely with senior engineers and product teams to deliver **high-quality, secure software**.

Technical Skills

Languages	Java	Kotlin	Python	TypeScript					
Backend & Frameworks	Spring	Boot	Spring	WebFlux	Django	REST Framework	REST	gRPC	Next.js
Messaging & Streaming	Kafka	RabbitMQ	AMQP						
Distributed Systems	Microservices		Event-Driven Design		Idempotency		Eventual Consistency		
	Fault Tolerance		High Availability						
Databases	PostgreSQL	MySQL	MS SQL	MongoDB	Redis	Flyway	Liquibase		
Cloud-Native Security	OAuth 2.0		Secrets Management		Encryption At Rest/In Transit				
DevOps & Cloud	Docker	Kubernetes	AWS	Azure	GitHub Actions		ArgoCD		
Testing & Quality	JUnit 5	Mockito	Testcontainers		TDD	Performance Testing			
Observability	Sentry	Proactive Monitoring			Incident Response		Logging		
Practices	Code Review		Documentation		Agile				

Key Capabilities & Impact

Capability	Impact
Production Java/Spring Boot at scale	5+ years shipping Spring Boot microservices powering digital banking, KYC, wallet and payment workflows at Paycloud Africa.
Microservices & event-driven architecture	Architected high-throughput REST APIs and asynchronous, event-driven microservices for real-time payment processing.
Event streaming & asynchronous messaging (Kafka/RabbitMQ)	Specialized in Kafka for event streaming and RabbitMQ/AMQP for reliable, async inter-service communication across distributed systems.
Idempotency, eventual consistency & fault tolerance	Designed idempotent processing and retry mechanisms that eliminated duplicate-transaction risk and cut payment-workflow bottlenecks.
Relational data modeling	Built and optimized PostgreSQL/MySQL-backed services across loan, payment and ERP integration systems.
Caching & containerization	Used Redis for caching and queue optimization, and Docker for consistent, containerized deployments.
Cloud infrastructure & CI/CD	Automated deployments and CI/CD on Azure-hosted infrastructure using GitHub Actions and ArgoCD.
Test-driven reliability	Safeguarded payment and onboarding workflows with unit, integration and performance tests using JUnit and Mockito.

Work Experience

Software Engineer

01/2024 – Present

Paycloud Africa Limited · Java, Spring Boot, RabbitMQ, Redis, MySQL, Next.js, Docker, K8s

- Architect **event-driven microservices** in Java and Spring Boot supporting digital banking, merchant onboarding, KYC workflows, wallet operations, and payment processing.
- Build high-throughput **RESTful APIs** handling financial transactions and real-time payment workflows, designed for **high availability and resilience** under load.
- Implement **asynchronous, event-driven communication** between microservices using **RabbitMQ and AMQP**, specializing in messaging and distributed systems.
- Design resilient microservices with **idempotent processing, retry mechanisms and fault-tolerant** patterns for asynchronous payment workflows, preserving **data integrity**.
- Write unit, integration and performance tests using **JUnit and Mockito** to ensure reliability of payment and onboarding workflows.
- Collaborate with DevOps, Product, QA and Ops to improve **CI/CD pipelines** via **GitHub Actions and ArgoCD**.

Software Engineer (Contract)

02/2025 – 10/2025

Digital Distribution Solutions · Kotlin, Spring Boot, Redis, Odoo, SAP, M-Pesa APIs, Docker

- Designed and implemented **ERP integrations** (Odoo & SAP) to unify order placement, automate sales logistics and digitize payment collection via M-Pesa.
- Introduced a cashless payment collection module, improving **transaction reliability and high availability** of reconciliation workflows.
- Built scalable backend services in **Kotlin + Spring Boot**, with proactive monitoring across key service endpoints.
- Reduced reconciliation errors and gave real-time visibility across sales and inventory, working with cross-functional Product and Ops teams.

Fullstack Developer

02/2022 – 12/2023

LipaLater · Java, Spring Boot, RabbitMQ, Redis, MySQL, Next.js, Docker, K8s

- Built scalable loan management services using **Java and Spring Boot microservices**, optimizing performance and reliability across distributed workflows.
- Designed **RESTful APIs** for loan allocation, repayment scheduling and customer lifecycle management.
- Developed an online loan application portal using Next.js, improving borrower onboarding.
- Optimized backend services using **asynchronous processing patterns** with **RabbitMQ and Redis**, improving loan processing efficiency by 30%.
- Implemented containerized deployments with Docker and GitHub Actions to streamline CI/CD.

Back End Developer

09/2020 – 02/2022

Konza Silicon Company · Python, Django, Redis, PostgreSQL, Nginx

- Developed **REST APIs** for authentication and land administration microservices using Python and Django REST Framework.
- Configured and managed Nginx for load balancing, reverse proxy and caching to support **high-availability** services.
- Contributed to the Ardhisasa land digitization platform; mentored junior team members through debugging and code reviews.

Education

Bachelor of Science, Mathematics and Computer Science

Jomo Kenyatta University of Agriculture and Technology, Kenya

References

James Mwai

CTO & Co-Founder, Paycloud Africa · james@paycloudafrica.com